

Notes: Ambuehl & Bernheim (AER, 2026)

Social Preferences over Ordinal Outcomes

Contents

1	Big picture	3
2	Conceptual framework and key objects	4
2.1	Planner, Stakeholders, and ordinal outcomes	4
2.2	Social choice rules as social preferences	4
2.3	Scoring rules	4
2.4	Concavity and compromise	5
2.5	Condorcet and supermajority rules	5
2.6	Runoff and multistage rules	5
3	Experimental design and data	6
3.1	Main work-domain experiment	6
3.2	Political-domain experiment	6
3.3	Choice profiles	6
3.4	Subject pool and implementation	6
3.5	Additional elicitations	7
4	Models and empirical specifications	7
4.1	Rule fingerprints	7
4.2	Bayesian classification	7
4.3	Random-choice benchmark	8
4.4	Goodness of fit	8
4.5	Out-of-sample prediction score	8
4.6	Testing cardinal inference	9
4.7	Testing intrinsic concern for ranks	9
5	Identification and interpretation	9
5.1	Why the design identifies ordinal aggregation preferences	9
5.2	What the classification can and cannot show	10
5.3	Why concavity matters	10
5.4	Evidence against pure majoritarianism	10
5.5	Evidence for structural stability	10
5.6	Evidence for both cardinal inference and intrinsic ranks	10

6	Tables and figures	11
6.1	Table 1: three-alternative profiles and rule predictions	11
6.2	Figure 1: rule fingerprints	11
6.3	Figure 2: decision interface	12
6.4	Figure 3: aggregate choice frequencies	12
6.5	Figure 4: best-fitting prespecified rules	12
6.6	Figure 5: empirical fingerprints of classified subjects	13
6.7	Figure 6: work versus political domain	13
6.8	Table 2: out-of-sample predictive power	13
6.9	Table 3: best-fit scoring parameters and cardinal beliefs	14
6.10	Table 4: information treatments	14
6.11	Table 5: cardinal payoff profiles	14
6.12	Table 6: choices with cardinal payoff information	16
6.13	Figure 7: general population samples	16
7	Short summary	16
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Takeaway	17

1 Big picture

- **Question.** How do people aggregate other people's *ordinal* preferences when they must choose for a group but do not have reliable cardinal utility information? In other words, what social choice principles do people use when they know only ranks?
- **Setting.** Subjects called **Planners** make consequential choices for groups of five **Stakeholders**. The main experiment uses two domains: assigning work tasks to workers and directing political contributions to parties. Supplemental experiments use charity-allocation choices in the United States and Sweden.
- **Core conceptual challenge.** There are astronomically many possible mappings from preference profiles to social choices. The paper therefore interprets observed choices through a structured menu of familiar social choice rules, then checks whether this menu omits important patterns.
- **Main contribution.** The paper empirically studies *social preferences over ordinal outcomes*. This differs from the usual Bergson-Samuelson approach, which assumes the social planner aggregates cardinal utilities.
- **Main empirical object.** A social choice rule maps a profile of Stakeholder rankings into a set of socially best alternatives. A Planner's observed choices are treated as revealing the rule that best summarizes her aggregation preferences.
- **Main rule family.** Most subjects behave as if they use **scoring rules**: assign a score to each rank, sum the scores across Stakeholders, and choose the alternative with the highest total score.
- **Main baseline result.** A clear majority of subjects are classified as using **strictly concave scoring rules**: they put more weight on moving someone out of last place than on moving someone into first place. In the paper's main classification, 62.5% of subjects are assigned to strictly concave scoring rules.
- **Most common rules.** The two most common fitted rules are the **Borda rule** and a **near-antiplurality rule**. Together they account for over 40% of subjects.
- **Majoritarian rules are uncommon.** Plurality, Condorcet, supermajority, unanimity, and runoff rules explain relatively few subjects. This is important because many institutions are built around majoritarian or strategic-voting criteria.
- **Stability result.** Classifications predict choices well out of sample and even across domains. Social preferences over ordinal outcomes therefore appear to contain stable structural elements, though there is some domain dependence.
- **Mechanism result.** Ranks matter in two ways. First, Planners use ranks to infer cardinal utility. Second, Planners also care about ordinal rankings intrinsically, even when cardinal payoff information is directly provided.
- **External-validity result.** General-population samples in the United States and Sweden display very similar distributions of aggregation principles, despite different political and social traditions.

- **Economic takeaway.** Many people dislike outcomes that leave some group members with their least preferred option. This gives ordinal social preferences a strong compromise flavor and helps explain why normatively appealing rules may be vulnerable to manipulation, while incentive-compatible rules may feel unattractive.

2 Conceptual framework and key objects

2.1 Planner, Stakeholders, and ordinal outcomes

- A Planner chooses one social alternative from a set of K alternatives, denoted $\mathcal{A}$.
- There are N Stakeholders. Each Stakeholder i has a strict ranking $\succ_i$ over $\mathcal{A}$.
- The Planner observes the ordinal preference profile

$$P = (\succ_1, \dots, \succ_N).$$

- The Planner's choice affects Stakeholders but not the Planner herself.

Interpretation. The experiment tries to isolate disinterested social aggregation: the Planner cannot benefit personally, cannot impose her own taste over the alternatives, and only sees Stakeholders' rankings.

2.2 Social choice rules as social preferences

A social choice rule is a mapping

$$R : P \mapsto R(P) \subseteq \mathcal{A}, \quad R(P) \neq \emptyset.$$

If $R(P)$ contains one alternative, the rule is **resolute**. If it contains more than one alternative, the rule is **irresolute**.

Interpretation. The paper does not treat a social choice rule as a literal voting procedure used by the subjects. It treats the rule as an *as-if representation* of the Planner's normative judgment, much like a utility function represents individual preference.

2.3 Scoring rules

For the main choice problems, there are three alternatives and five Stakeholders. A scoring rule assigns scores to ranks:

$$f_s(1) = 1, \quad f_s(2) = s, \quad f_s(3) = 0, \quad s \in [0, 1].$$

For an alternative x , let $r_i(x)$ be the rank assigned by Stakeholder i . The total score is

$$S_s(x) = \sum_{i=1}^N f_s(r_i(x)),$$

and the scoring rule selects

$$R_s(P) = \arg \max_{x \in \mathcal{A}} S_s(x).$$

Interpretation. The single parameter s tells us how much the Planner values a second-place outcome relative to a first-place and third-place outcome.

2.4 Concavity and compromise

- If $s < 1/2$, the scoring rule is **convex**. It places more weight on creating first-place outcomes.
- If $s = 1/2$, the scoring rule is **linear**. This is the Borda rule.
- If $s > 1/2$, the scoring rule is **concave**. It places more weight on avoiding third-place outcomes.
- If $s = 0$, the rule resembles plurality: count first-place ranks.
- If $s = 1$, the rule resembles antiplurality: avoid last-place ranks.

Interpretation. Concavity is the paper’s central economic idea. A concave scoring rule says that moving a Stakeholder from worst to middle matters more than moving a Stakeholder from middle to best. This is a preference for compromise and last-place avoidance.

2.5 Condorcet and supermajority rules

Condorcet-style rules compare alternatives pairwise. Alternative A is socially preferred to B if the fraction of Stakeholders who prefer A to B is at least q , where

$$q \geq \frac{1}{2}.$$

The special case $q = 1/2$ is the Condorcet relation; $q = 1$ is unanimity.

Interpretation. These rules emphasize pairwise majority support. The paper’s results show that such majoritarian logic describes relatively few Planners.

2.6 Runoff and multistage rules

The paper also considers scoring runoff rules. These repeatedly eliminate the alternative with the lowest score until one alternative remains.

Interpretation. Runoff rules introduce a procedural flavor. They are important because many real elections use multistage procedures, but they are not the dominant explanation of subjects’ choices.

3 Experimental design and data

3.1 Main work-domain experiment

- Stakeholders are Amazon Mechanical Turk workers.
- Each worker receives \$15 for completing one assigned task.
- The five possible work tasks include image labeling, hate speech filtering, audio transcription, movie review classification, and a logic-based task.
- Workers first reveal their preferences over tasks.
- Planners then choose among task assignments based only on ordinal rankings of assignments.

Interpretation. The work-domain design makes choices consequential but prevents Planners from substituting their own views about which task is good or bad.

3.2 Political-domain experiment

- Stakeholders are voting-age Swiss citizens.
- Planners direct a contribution of SFr30 to one of the five largest Swiss political parties.
- Planners observe rankings over anonymous symbols, not the identities of the parties.

Interpretation. The anonymity of the parties matters. It prevents the Planner from using her own political ideology and keeps the focus on aggregation of others' rankings.

3.3 Choice profiles

- The main work-domain analysis uses 17 discerning preference profiles with five Stakeholders and three alternatives.
- These profiles are chosen because they separate social choice rules rather than because they are naturally common.
- The political-domain experiment uses a smaller set of seven starred profiles from the same profile list.

Interpretation. The profile selection is like an experimental identification design. The profiles are chosen to make different rules produce different observable choices.

3.4 Subject pool and implementation

- The main analysis uses 405 Planner subjects.

- Subjects are Swiss citizens, mainly students at the University of Zurich and the Swiss Federal Institute of Technology.
- Subjects complete both work-domain and political-domain decisions, with randomized domain order and randomized profile order.
- One out of four Planners is matched to a real Stakeholder group, and one decision may be implemented.

Interpretation. The random implementation design gives Planners an incentive to treat each choice as potentially consequential for real people.

3.5 Additional elicitations

The paper also measures:

- Planners' beliefs about Stakeholders' reservation wages or willingness to pay.
- Risk preferences using the Holt-Laury method.
- Altruism using modified dictator games.
- Cognitive reflection and knowledge of social choice theory.

Interpretation. These additional measures help distinguish whether ranks matter because they proxy for cardinal utility or because people intrinsically care about rank positions.

4 Models and empirical specifications

4.1 Rule fingerprints

Each social choice rule produces a distinctive **fingerprint**: for every profile, it predicts a set of best alternatives. For example, a scoring rule with parameter s predicts different choices as s crosses thresholds such as $1/3$, $1/2$, $3/5$, $2/3$, $3/4$, $4/5$, and 1 .

Interpretation. Fingerprints convert an abstract rule into an empirical object. A subject's observed choices can be compared to the fingerprints of many candidate rules.

4.2 Bayesian classification

Let c_i denote a subject's observed choice in profile i . For a candidate rule R and noise parameter $\varepsilon \in [0, 1]$, the paper assumes the subject follows the rule with probability $1 - \varepsilon$ and chooses randomly with probability ε .

If the choice set prescribed by the rule is $R(P_i)$, then the likelihood contribution is

$$\Pr(c_i | R, \varepsilon) = \begin{cases} \frac{1 - \varepsilon}{|R(P_i)|} + \frac{\varepsilon}{3}, & c_i \in R(P_i), \\ \frac{\varepsilon}{3}, & c_i \notin R(P_i). \end{cases}$$

Assuming independent errors across profiles, the likelihood is

$$L(R, \varepsilon | c) = \prod_i \Pr(c_i | R, \varepsilon).$$

With uniform priors over rules and ε , the subject is assigned by maximum a posteriori classification:

$$(R^*, \varepsilon^*) = \arg \max_{R, \varepsilon} P(R, \varepsilon | c).$$

Interpretation. Irresolute rules do not get full credit whenever they include the observed choice. If a rule says two alternatives are tied, choosing one of them is less diagnostic than if the rule uniquely predicts it.

4.3 Random-choice benchmark

To judge whether observed classifications are meaningful, the paper simulates artificial subjects who choose uniformly at random across the three options. It then applies the same Bayes classifier to these simulated subjects.

Interpretation. This benchmark matters because a flexible classifier will always find some rule that fits random data. The paper asks whether the observed fraction assigned to each rule is unusually high or low relative to this random baseline.

4.4 Goodness of fit

The estimated noise parameter ε^* summarizes how often a subject's choices behave as if they deviate from the assigned rule. In the main analysis, the average estimated noise parameter is 0.10, and 42.0% of subjects fit their assigned rule perfectly.

Interpretation. The high fit is important because the paper is not merely forcing noisy behavior into arbitrary categories. Many subjects' full choice vectors closely match recognizable social choice rules.

4.5 Out-of-sample prediction score

For prediction exercises, the rule assigns a predicted best-choice set. If the actual choice is outside the set, the score is zero. If it is inside the set and the set contains m alternatives, the score is

$$\text{score} = \frac{1}{m}, \quad m \in \{1, 2, 3\}.$$

Uniform random choice yields expected score 1/3 regardless of the size of the predicted choice set.

Interpretation. This scoring method rewards sharper predictions. A rule that uniquely predicts the chosen option receives more credit than a rule that says all options are tied.

4.6 Testing cardinal inference

If Planners use ranks only to infer cardinal values, then their best-fit scoring parameter should be related to their beliefs about the cardinal value of ranks. In the work domain, let u_r denote the average dollar-equivalent reservation wage associated with a Stakeholder’s r -th ranked task. A money-metric utilitarian objective is equivalent to a scoring rule with synthetic parameter

$$\tilde{s} = \frac{u_3 - u_2}{u_3 - u_1}.$$

The paper estimates regressions relating best-fit s to $\tilde{s}$, risk tolerance, altruism, domain indicators, and presentation controls.

Interpretation. A positive relationship between s and $\tilde{s}$ supports the idea that subjects partly use ranks to infer cardinal utility.

4.7 Testing intrinsic concern for ranks

To test whether ranks matter intrinsically, the paper runs treatments in which Planners observe actual monetary payoffs. These treatments change ordinal ranks while holding cardinal payoff distributions fixed, or make one option cardinally dominant while changing its rank profile.

Interpretation. If choices still react to rank profiles when cardinal payoffs are known, then ordinal information is not merely a proxy for cardinal utility. It is part of the social objective.

5 Identification and interpretation

5.1 Why the design identifies ordinal aggregation preferences

- **No self-interest.** The Planner’s own payoff does not depend on which social alternative she selects.
- **No paternalistic override.** The Planner sees anonymous symbols and rankings, not the substantive identity of tasks, assignments, parties, or charities.
- **Consequential choices.** Choices may be implemented for real Stakeholders, so they are not merely hypothetical.
- **Incentive-compatible Stakeholder rankings.** Workers reveal task rankings under procedures designed to reduce strategic reporting.
- **Rule-separating profiles.** The 17 main profiles are selected to generate different predictions across candidate rules.

Interpretation. The experiment strips away many confounds that usually make social choice data hard to interpret. What remains is the Planner’s view about how to aggregate ranks.

5.2 What the classification can and cannot show

The Bayes classifier does not prove that subjects consciously compute Borda or antiplurality scores. It shows that their choices are *as if* they followed those rules.

Interpretation. The right analogy is demand theory: a consumer may not calculate a utility function, but her choices can still be represented as maximizing one.

5.3 Why concavity matters

Concave scoring rules are not the same as Rawlsian maximin or cardinal inequality aversion. They are based on *within-menu ranks*. A person is considered badly treated if she receives her least preferred alternative in the current menu, even if cardinally the loss may be small.

Interpretation. The paper’s compromise result is ordinal. It is about avoiding last-place outcomes according to each Stakeholder’s own ranking.

5.4 Evidence against pure majoritarianism

The data show low prevalence of Condorcet and plurality logic. This is striking because those rules are familiar and often central to voting theory.

Interpretation. People may prefer outcomes that are not majority winners if those outcomes avoid making some Stakeholders particularly badly off in ordinal terms.

5.5 Evidence for structural stability

The same classification predicts choices across profiles and across domains, but not perfectly. The political domain shifts subjects somewhat away from high concavity and toward Borda or plurality-equivalent rules.

Interpretation. Social preferences over ordinal outcomes are not entirely context-free, but they are also not arbitrary. There is a stable underlying preference for compromise.

5.6 Evidence for both cardinal inference and intrinsic ranks

The paper’s mechanism evidence has two parts. The correlation between $\tilde{s}$ and best-fit s supports cardinal inference. The payoff treatments, where monetary payoffs are known, show intrinsic concern for rank profiles.

Interpretation. Ranks are not just noisy labels for utility. They also carry direct normative weight.

6 Tables and figures

6.1 Table 1: three-alternative profiles and rule predictions

Read. Table 1 lists the 17 core preference profiles and the predictions of selected aggregation rules. It shows how different values of the scoring parameter s , runoff rules, and Condorcet-style rules imply different best choices.

Economic message. The table is the identification backbone of the paper. The experiment is designed so that a subject's choices across these profiles reveal whether she behaves like a plurality, Borda, antiplurality, Condorcet, runoff, or other rule user.

Rule predictions													
Index (1)	Profile (2)	Scoring cutoff	Scoring rules, $s \in$						Runoff cutoff	Runoff rules, $s \in$			Condorcet (13)
		$\bar{s}$ (3)	$\{0\}$ (4)	$\{0,\bar{s}\}$ (5)	$\{\bar{s}\}$ (6)	$\{\bar{s},1\}$ (7)	$\{1\}$ (8)	$\hat{s}$ (9)	$[0,\hat{s}]$ (10)	$\{\hat{s}\}$ (11)	$[\hat{s},1]$ (12)		
1*	B B A A A C C B B B A A C C C	1/3	A	A		{A,B}	B	B	–	A	A	A	A
2	B C C A A A B B B B C A A C C	1/3	{A,C}	A		{A,B}	B	B	1/4	A	A	B	B
3*	A A A B B C B B C C B C C A A	1/2	A	A		{A,B}	B	B	1	A	B	–	A
4*	B B A A A A C B B B C A C C C	1/2	A	A		{A,B}	B	B	–	A	A	A	A
5	C A A A B B B B B C A C C C A	1/2	A	A		{A,B}	B	B	–	A	A	A	A
6*	C C A A A B B B B B A A C C C	3/5	A	A		{A,B}	B	B	–	A	A	A	A
7	C A A A B B C B B C A B C C A	2/3	A	A		{A,B}	B	B	1	A	B	–	A
8*	C A A A B B B B B A A C C C C	2/3	A	A		{A,B}	B	B	–	A	A	A	A
9*	A A A C C C B B B B B C C A A	3/4	A	A		{A,B}	B	B	1	A	B	–	A
10	B A A A A C B B B B A C C C C	3/4	A	A		{A,B}	B	B	–	A	A	A	A
11*	C A A A A B B B B B A C C C C	4/5	A	A		{A,B}	B	B	–	A	A	A	A
12	C B B A A A C C B B B A A C C	0	–	–		{A,B}	B	B	1	A	B	–	{A,B,C}
13	B C C A A A B B C B C A A B C	1/2	{A,C}	{A,C}		{A,B,C}	B	B	1/2	A	{A,B,C}	B	{A,B,C}
14	C B B A A A C A B B B A C C C	–	{A,B}	{A,B}		{A,B}	{A,B}	{A,B}	–	A	A	A	A
15	B C C A A A B A B B C A B C C	0	–	–		{A,C}	A	{A,B}	–	A	A	A	A
16	C B B A A A C A B C B A C C B	0	–	–		{A,B}	A	A	–	A	A	A	A
17	B C C A A A A A B B C B B C C	0	–	–		{A,C}	A	A	–	A	A	A	A

Notes: Each profile is displayed as a 3×5 matrix. Columns correspond to Stakeholders and rows to preference ranks. A Stakeholder's first-, second-, and third-ranked alternatives are listed in the first, second, and third rows, respectively. For Condorcet-cyclical profiles, we indicate the set of options in the top cycle. For decisions in the political domain, we only use the profiles indicated with an asterisk.

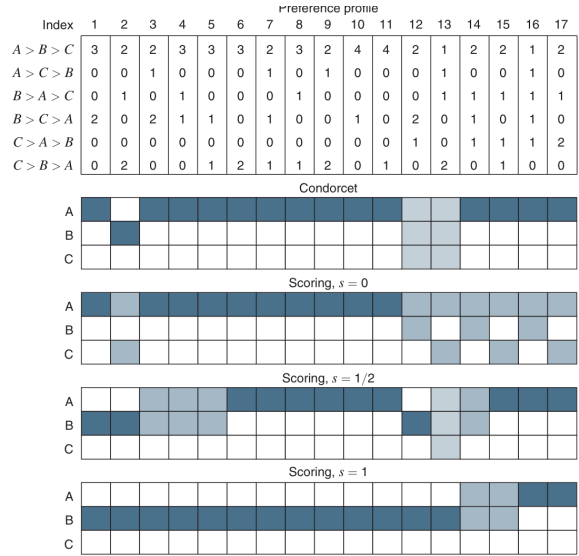


FIGURE 1. FINGERPRINTS OF AN EXAMPLE SELECTION OF ORDINAL AGGREGATION RULES

Notes: Each column corresponds to a three-alternative preference profile. The profiles appear in the top panel: Each cell shows the number of Stakeholders, with the preference ranking indicated to the left of the row. In the second through fifth panels, we use dark shading if the rule chooses the corresponding option and no shading if the rule does not choose the option. Nonresolute rules choose more than one option for some profiles. We use intermediate shades to indicate the number of tied options.

6.2 Figure 1: rule fingerprints

Read. Figure 1 visualizes the fingerprints of several aggregation rules. Each column is a preference profile and each row shows whether a rule selects alternative A , B , or C .

Economic message. The figure explains the classifier's logic. Rules are not identified from one choice, but from a pattern across many rule-separating profiles.

6.3 Figure 2: decision interface

Read. Figure 2 shows the interface Planners use. Vertical bars represent workers, and symbols represent assignments. The Planner can highlight or hide alternatives and then choose an assignment for the group.

Economic message. The interface makes ordinal information visible while hiding the substantive content of the assignments. That is crucial for isolating aggregation preferences from paternalistic judgments.

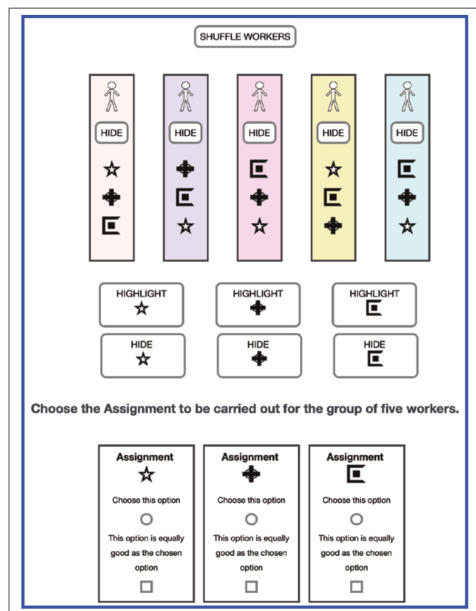


FIGURE 2. PLANNERS' DECISION INTERFACE

Notes: Subjects can drag and drop columns, click a button to shuffle columns, highlight choice options, hide choice options, and hide workers. If a subject hides an option, the remaining options are arranged on two rows regardless of their initial position. Symbols for options and colors for the bars representing each worker are randomly drawn each round. The order of workers is randomly drawn each round.

Profile	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Condorcet scoring	A	B	A	A	A	A	A	A	A	A	A	ABC	ABC	A	A	A	A
$\bar{s}$	1/3	1/3	1/2	1/2	1/2	3/5	2/3	2/3	3/4	3/4	4/5	0	1/2	-	0	0	0
$s < \bar{s}$	A	A ^(a)	A	A	A	A	A	A	A	A	A	- ^(b)	AC	AB	- ^(a)	- ^(b)	- ^(a)
$s > \bar{s}$	B	B	B	B	B	B	B	B	B	B	B	B	B	AB	A ^(c)	A	A
Choice freq.																	
A	13	2	36	37	24	38	56	60	72	70	70	8	15	62	93	98	99
B	86	96	64	63	75	61	43	39	25	29	29	91	73	37	6	1	1
C	0	1	0	0	1	0	1	1	3	0	1	1	13	1	1	1	0

FIGURE 3. DISTRIBUTION OF CHOICES IN THREE-OPTION PROBLEMS

Notes: (a) For $s = 0$, $\{A, C\}$ is selected. (b) For $s = 0$, $\{A, B\}$ is selected. (c) For $s = 1$, $\{A, B\}$ is selected. Darker shades of blue indicate higher frequencies. The numerical percentage appears within each shaded cell. The profile numbering is the same as for Table 1.

6.4 Figure 3: aggregate choice frequencies

Read. Figure 3 reports the distribution of choices across the 17 three-option work-domain profiles. The rows list choices A , B , and C , and darker cells indicate higher frequencies.

Economic message. Choices are highly structured. Many subjects select alternatives consistent with concave scoring rules, and very few select alternatives consistent with extreme convexity or simple majoritarian logic.

6.5 Figure 4: best-fitting prespecified rules

Read. Figure 4 presents the main classification results. The largest bars are for Borda and near antiplurality, and most mass lies on concave scoring rules. Red dots and whiskers show the random-choice benchmark.

Economic message. The central result is that most subjects put high value on avoiding worst-ranked outcomes. Plurality, supermajority, unanimity, and runoff rules are not empirically dominant.

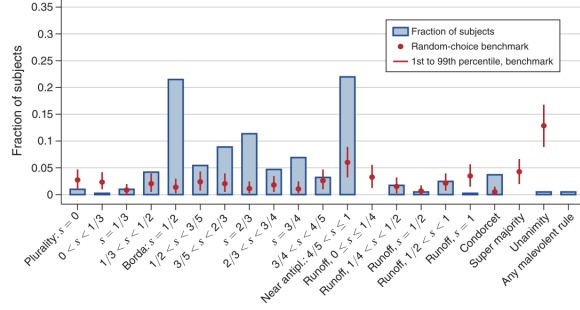


FIGURE 4. BEST-FITTING PRESPECIFIED RULES

Note: The random-choice benchmark for malevolent rules exceeds the range of the graph (44.8 percent).

Panel A. Subjects classified as Borda ($s = 0.5$)

	Empirical choices																
A	8	0	66	55	40	84	94	95	95	95	97	1	28	66	99	98	100
B	92	99	34	45	60	16	5	5	5	5	3	99	48	34	1	2	0
C	0	1	0	0	0	0	1	0	0	0	0	0	24	0	0	0	0

	Rule prediction																
A	0	0	50	50	50	100	100	100	100	100	100	0	33	50	100	100	100
B	100	100	50	50	50	0	0	0	0	0	0	100	33	50	0	0	0
C	0	0	0	0	0	0	0	0	0	0	0	0	33	0	0	0	0

Panel B. Subjects classified as near antiplurality ($4/5 < s < 1$)

	Empirical choices																
A	5	0	10	14	1	1	15	11	33	14	0	4	3	63	100	100	100
B	95	100	90	86	96	99	85	89	62	86	100	96	96	37	0	0	0
C	0	0	0	0	3	0	0	0	5	0	0	0	1	0	0	0	0

	Rule prediction																
A	0	0	0	0	0	0	0	0	0	0	0	0	0	50	100	100	100
B	100	100	100	100	100	100	100	100	100	100	100	100	100	50	0	0	0
C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FIGURE 5. EMPIRICAL FINGERPRINTS OF CLASSIFIED SUBJECTS

Notes: In both panels, the top half displays the choices by subjects assigned to a certain rule. The bottom half displays the choices the rule predicts. Darker shades of gray indicate higher frequencies. The numerical percentage appears within each shaded cell. For the theoretical rule, we take the frequency of each best option to be 50 percent for a 2-way tie and 33 percent for a 3-way tie.

6.6 Figure 5: empirical fingerprints of classified subjects

Read. Figure 5 compares the observed choices of subjects classified as Borda or near antiplurality with the theoretical predictions of those rules.

Economic message. The fit is not superficial. Subjects assigned to a rule make choices that closely resemble the full rule fingerprint across profiles.

6.7 Figure 6: work versus political domain

Read. Figure 6 plots the difference in classification frequencies between the political and work domains. Positive values mean the rule is more common in the political domain.

Economic message. There is domain dependence, but it is limited. The political domain is somewhat less concave, yet the broader compromise pattern remains.

6.8 Table 2: out-of-sample predictive power

Read. Table 2 reports prediction scores for classifications trained and tested within or across domains. Prediction scores range from 0.755 to 0.853, well above the random-choice benchmark of 0.333 and above random assignments based on estimated rule frequencies.

Economic message. The classification is not just in-sample curve fitting. It captures stable aggregation principles that predict new choices.

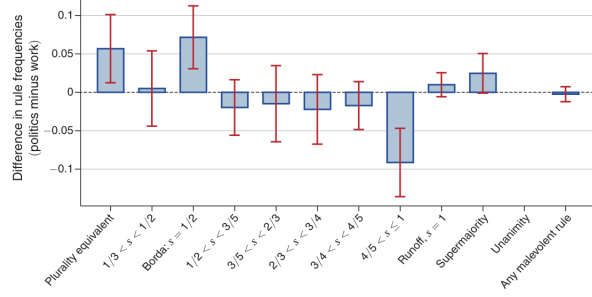


FIGURE 6. COMPARISON BETWEEN WORK AND POLITICAL DOMAINS

Notes: Classifications are based on the same set of seven preference profiles for both domains. The category “plurality equivalent” includes scoring rules with $s < 1/3$, the Condorcet rule, and all scoring runoff rules with $s < 1$. Whiskers denote 95 percent confidence intervals.

TABLE 2—OUT-OF-SAMPLE PREDICTIVE POWER OF THE BAYESIAN CLASSIFICATION

Test domain	Fraction of correct predictions (weighted by resoluteness)	
	Work (1)	Politics (2)
<i>Predictions</i>		
Training domain		
Work	0.853	0.758
Politics	0.805	0.755
<i>Benchmarks</i>		
1. Theoretical maximum	0.924	0.955
2. Random choice	0.333	0.333
3a. Estimated rule frequencies, work domain		
Mean	0.736	0.617
Ninety-ninth percentile	0.754	0.645
3b. Estimated rule frequencies, political domain		
Mean	0.716	0.642
Ninety-ninth percentile	0.736	0.669

Note: Within-domain predictions are based on a leave-out-one approach.

6.9 Table 3: best-fit scoring parameters and cardinal beliefs

Read. Table 3 relates best-fit scoring parameters to synthetic scoring parameters derived from Planners’ beliefs about reservation prices. The coefficient on the synthetic scoring parameter is positive and statistically significant.

Economic message. Part of the reason ranks matter is informational: Planners use ordinal ranks to infer cardinal utility intensity.

6.10 Table 4: information treatments

Read. Table 4 compares baseline profiles with profiles that add information about an unavailable option X and profiles that add X as an available option. Choice distributions shift substantially when X changes the information conveyed by ranks.

Economic message. Planners use rankings of unavailable alternatives to infer cardinal information. This supports the cardinal-inference mechanism.

6.11 Table 5: cardinal payoff profiles

Read. Table 5 displays monetary payoff profiles and the ordinal rankings they imply. It shows swap and addition treatments designed to separate cardinal-payoff concerns from ordinal-rank concerns.

Economic message. These profiles create clean tests of whether ranks matter after cardinal payoffs are made explicit.

TABLE 3—RELATION BETWEEN BEST-FITTING SCORING PARAMETERS AND BELIEFS ABOUT RESERVATION PRICES

Variable	Estimated scoring parameter s			
	(1)	(2)	(3)	(4)
<i>Scoring parameter $\bar{s}$</i>				
Level	0.115 (0.056)		0.102 (0.056)	
%-rank		0.062 (0.023)		0.055 (0.023)
<i>Risk tolerance</i>				
Level			−0.717 (0.314)	
%-rank				−0.054 (0.028)
<i>Altruism</i>				
Level			0.178 (0.223)	
%-rank				0.048 (0.028)
Political domain	−0.111 (0.010)	−0.111 (0.010)	−0.111 (0.010)	−0.112 (0.010)
Observations	792	792	792	792
Subjects	405	405	405	405

Notes: Parameters estimated with OLS using midpoint values of best-fit score. Regressions include all subjects with monotonic beliefs about reservation prices. Risk tolerance is measured as the certainty equivalent of an equiprobable win 40/win 10 lottery implied by individual-level CRRA estimates. Altruism is measured as the maximum amount of Swiss francs the Planner is willing to give up to increase each Stakeholder's payoff by \$2. For subjects with multiple switches in the risk preference or altruism elicitation, the corresponding variable is set to the mean value among the other subjects. We also include two indicator variables for the presence of multiple switching points, one for each characteristic. All regressions control for the type of preference profile presentation and for whether the political domain was displayed before or after the work domain. Standard errors are clustered by subject.

TABLE 5—CARDINAL PAYOFF PROFILES

(1)

Panel A. Swap treatments

S1: Original cardinal payoff profile

Option	Stakeholder				Preference	Stakeholder		
	1	2	3			1	2	3
A	6	3	5	swap ↔ swap	Best	A	B	B
B	3	5	6		Middle	B	C	A
C	1	4	2		Worst	C	A	C

S1': Altered cardinal payoff profile

Option	Stakeholder				Preference	Stakeholder		
	1	2	3			1	2	3
A	6	5	3		Best	A	A	B
B	5	3	6		Middle	B	B	A
C	4	1	2		Worst	C	C	C

(2)

Panel B. Addition treatments

A1: Original cardinal payoff profile

Option	Stakeholder				Pref. rank	Stakeholder		
	1	2	3			1	2	3
A	3	2	2		Best	A	B	B
B	1	3	3		Worst	B	A	A

A1': Altered cardinal payoff profile

Option	Stakeholder				Preference	Stakeholder		
	1	2	3			1	2	3
A	3	2	2		Best	A	B	B
B	1	3	3		Middle	X	A	A
X	2	1	1		Worst	B	X	X

TABLE 4—INFORMATION TREATMENTS

(1)				(2)			(3)		
Panel A. Baseline profiles									
Profiles	B B A A A A A B B B			B A A A B A B B B A			A A B B C B B A C A C C C A B		
Choice distribution	A	B		A	B		A	B	C
	0.993	0.007		0.993	0.007		0.622	0.368	0.010
Panel B. Adding information about an unavailable option X									
Profiles	B B A A A A X B B B X A X X X			X A A A B B B B B X A X X X A			A A B B C B B A C X C C C X A X X X A B		
Choice distribution	A	B		A	B		A	B	C
	0.427	0.573		0.358	0.642		0.217	0.674	0.109
Panel C. Adding an available option X									
Profiles	B B A A A A X B B B X A X X X			X A A A B B B B B X A X X X A			A A B B C B B A C X C C C X A X X X A B		
Choice distribution	A	B	X	A	B	X	A	B	X
	0.365	0.630	0.005	0.240	0.748	0.012	0.205	0.694	0.099
								0.002	
Panel D. p-values of the difference of choice distributions									
Comparison across panels									
A and B	0.000			0.000			0.000		
A and C	0.000			0.000			0.000		
B and C	0.531			0.021			1.000		

Notes: Column 3 of panel A is profile 14 of Table 1. In panel C, column 3 is profile 18 from Supplemental Appendix Table B.1, and columns 1 and 2 are profiles 4 and 5 from Table 1, respectively. Columns 1 and 2 of panel A both display the distribution of choices that subjects made for the single two-alternative profile they encountered. All decisions concern the work domain. p -values are based on two-sample Kolmogorov-Smirnov tests for equality of distributions among options other than X.

TABLE 6—CHOICES WITH CARDINAL PAYOFF INFORMATION

	Option A chosen		
Profile	S1	S1'	Difference
<i>Panel A. Swap treatments</i>			
$\epsilon = 0$	0.300 (0.013)	0.712 (0.012)	0.412 (0.022)
$\epsilon > 0$	0.755 (0.014)	0.901 (0.010)	0.146 (0.015)
Difference	0.455 (0.017)	0.189 (0.013)	−0.265 (0.021)
	Option B chosen		
Profile	A1	A2	
<i>Panel B. Addition treatments</i>			
X unavailable	0.220 (0.014)	0.496 (0.019)	
X added	0.175 (0.013)	0.396 (0.018)	
Difference	0.045 (0.014)	0.099 (0.014)	

Notes: Panel A shows the frequency with which subjects choose option A. Panel B shows the frequency with which subjects choose option B. Standard errors clustered by subject. Pooled across display formats.

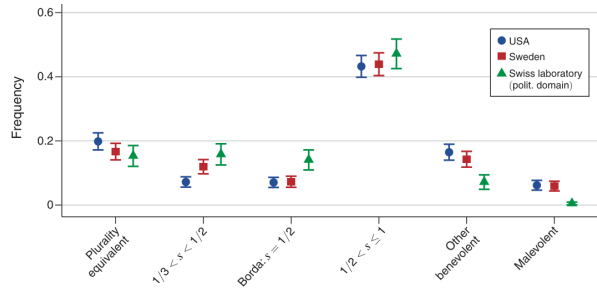


FIGURE 7. CLASSIFICATION OF GENERAL POPULATION SAMPLES TO PRESPECIFIED RULES

Notes: Bayes classifications based on consequential choices for the seven three-option preferences profiles indicated with stars in Table 1. General population observations weighted to make the samples representative with respect to gender and party preference. Whiskers denote 95 percent confidence intervals.

6.12 Table 6: choices with cardinal payoff information

Read. Table 6 shows that Planners' choices shift when ordinal rankings change, even when cardinal payoffs are known. In the swap treatment with $\epsilon = 0$, option A is chosen 0.300 in $S1$ but 0.712 in $S1'$. With $\epsilon > 0$, the corresponding frequencies are 0.755 and 0.901.

Economic message. The evidence rejects the view that ordinal ranks matter only as a proxy for cardinal utility. Planners intrinsically care about rank profiles.

6.13 Figure 7: general population samples

Read. Figure 7 classifies US and Swedish general-population subjects and compares them with the Swiss laboratory political-domain sample. The distributions are broadly similar.

Economic message. The preference for compromise and concave scoring is not unique to Swiss student subjects. It appears robust across countries with different political traditions.

7 Short summary

- The paper asks how people aggregate ordinal preferences when cardinal utility information is missing or unreliable.
- Subjects make consequential choices for groups while observing only the Stakeholders' rankings over anonymous alternatives.
- The main empirical method classifies each subject according to the social choice rule that best fits her choice fingerprint.
- Most subjects behave as if they use scoring rules, especially Borda and near antiplurality.
- A majority use strictly concave scoring rules, meaning they emphasize avoiding last-place outcomes rather than maximizing first-place outcomes.
- Majoritarian rules such as plurality and Condorcet explain relatively few subjects.

- Classifications fit well, survive random-choice benchmarks, and predict choices out of sample and across domains.
- Ranks matter partly because they help Planners infer cardinal utility, but also because people care about ordinal outcomes intrinsically.
- US and Swedish general-population samples show similar aggregation preferences.

8 Conclusion

8.1 Core conclusion

The paper shows that social preferences over ordinal outcomes are highly structured. Most people do not simply follow majority winners or first-place counts. Instead, they often prefer compromise outcomes that avoid giving some Stakeholders their worst-ranked option.

8.2 Economic intuition

The key intuition is last-place aversion in social aggregation. When choosing for a group, many people regard a middle-ranked outcome for everyone as more attractive than an outcome that gives some people their favorite option while leaving others with their least favorite option.

8.3 Takeaway

Ordinal aggregation is not only a technical social choice problem. It is also a behavioral and normative object. People have stable views about how rankings should be aggregated, and those views often favor concave scoring rules that are normatively appealing but may be strategically manipulable in practice.